

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Matematica



UNIVERSITÀ  
DI TORINO

Tesi di Laurea Magistrale

## The ideally exact categories

A non-pointed generalization  
of the notion of semiabelian category

**Relatore**

professor Alan Cigoli

**Candidato**

Christian D'Ascoli

Anno Accademico 2024-2025

# Table of Contents

|          |                                                   |          |
|----------|---------------------------------------------------|----------|
| <b>1</b> | <b>Theory</b>                                     | <b>3</b> |
| 1.1      | Category over an object . . . . .                 | 3        |
| 1.2      | Connected limits . . . . .                        | 3        |
| 1.3      | The pullback functor and its adjunction . . . . . | 5        |
| 1.4      | The monad associated to the adjunction . . . . .  | 11       |



# Introduction

Textt



# Chapter 1

## Theory

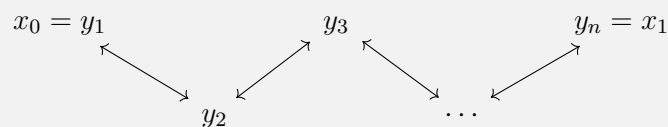
### 1.1 Category over an object

Let  $E$  be an object in a category  $\mathcal{A}$ .  $(\mathcal{A} \downarrow E)$  is the category of objects over  $E$  in  $\mathcal{A}$ , we denote an object by  $(A, \alpha : A \rightarrow E)$ .

A morphism between two objects  $(A, \alpha)$  and  $(A', \alpha')$  is a morphism  $f : A \rightarrow A'$  such that  $\alpha' f = \alpha$ .

### 1.2 Connected limits

**Definition 1.2.1.** A connected limit is a limit over a connected category. A category is connected if it is inhabited and any two objects can be connected by a zigzag of morphisms



where each symbol  $\leftrightarrow$  is a morphism  $y_i \rightarrow y_{i+1}$  or  $y_{i+1} \rightarrow y_i$ .

**Example 1.2.2.** Some examples

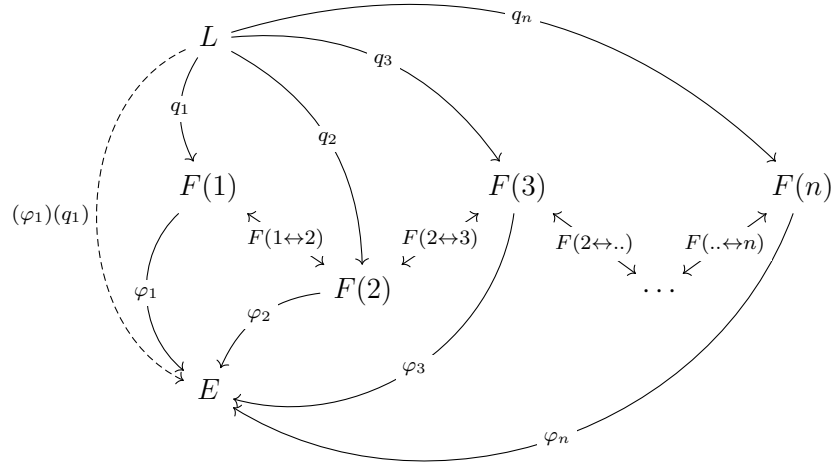
- The pullback is a connected limit.

$$\begin{array}{ccc}
 & & 2 \\
 & & \downarrow j \\
 1 & \xrightarrow{i} & 3
 \end{array}$$

- The equalizer is a connected limit.

$$1 \begin{array}{c} \xrightarrow{i} \\ \xrightarrow{j} \end{array} 2$$

**Remark 1.2.3.** The connected limits in the category  $(\mathcal{A} \downarrow E)$  can be computed in  $\mathcal{A}$ . In fact we can provide to the objects in  $\mathcal{A}$  a morphism to form an object in  $(\mathcal{A} \downarrow E)$ . In the zigzag below we chose for example  $(\varphi_1)(q_1)$ , but we could use whatever morphism of the type  $(\varphi_i)(q_i)$ . The connection guarantees that all the morphisms  $q_1 \dots q_n$  involved in the limit are also morphisms in  $(\mathcal{A} \downarrow E)$ , ie  $(\varphi_i)(q_i) = (\varphi_1)(q_1) \forall i \in 1, \dots, n$ .



**Remark 1.2.4.** In particular one can affirm that, for each object  $E$  in  $\mathcal{A}$ ,  $(\mathcal{A} \downarrow E)$  has all the connected limits that  $\mathcal{A}$  has.

**Example 1.2.5.** The product is not a connected limit, in fact we can see that the following category is not connected.

$$1 \quad 2$$

We try to compute the product directly in  $(\mathcal{A} \downarrow E)$ . Let  $(D, \delta)$  and  $(D', \delta')$  be two objects in  $(\mathcal{A} \downarrow E)$ , the product is given by the following pullback

$$\begin{array}{ccc} D \times_E D' & \xrightarrow{\quad} & D' \\ \swarrow & & \searrow \delta' \\ D & \xrightarrow{\delta} & E \\ \searrow \delta & & \parallel \\ & & E \end{array}$$

which doesn't correspond to the product  $D \times D'$  in  $\mathcal{A}$ .

## 1.3 The pullback functor and its adjunction

Given a category  $\mathcal{A}$  with pullbacks and a morphism  $p : E \rightarrow B$  in  $\mathcal{A}$ , one can consider functors pullback and shrink

$$\begin{array}{ccc}
 & p^* & \\
 (\mathcal{A} \downarrow B) & \xrightarrow{\quad} & (\mathcal{A} \downarrow E) \\
 & p_! &
 \end{array}$$

defined as follows:

- $p^* : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$  is the pullback functor

$$\begin{array}{ccc}
 \begin{array}{ccc} A & & \\ \downarrow \alpha & & \\ B & & \end{array} & \xrightarrow{p^*} & \begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A & & \\ \downarrow \text{proj}_1 & & \downarrow \alpha & & \\ E & \xrightarrow{p} & B & & \end{array}
 \end{array}$$
  

$$\begin{array}{ccc}
 \begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \nearrow \alpha' & \\ B & & \end{array} & \xrightarrow{p^*} & \begin{array}{ccccccc} & & \exists! \tilde{f} & & & & \\ & \xrightarrow{\quad} & \text{dashed} & \xrightarrow{\quad} & & & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & & \nearrow \alpha' & & & \\ & & B & & & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \searrow \text{proj}_1' & & \\ & E & & & & & \end{array}
 \end{array}$$

- $p_! : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$  is the shrink functor

$$\begin{array}{ccc}
 \begin{array}{ccc} D & & \\ \downarrow \delta & & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & & \\ \delta \downarrow & \searrow p\delta & \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$
  

$$\begin{array}{ccc}
 \begin{array}{ccc} D & \xrightarrow{d} & D' \\ \delta \downarrow & \nearrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{d} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

**Proposition 1.3.1.** *Let  $p : E \rightarrow B$  be a morphism in  $\mathcal{A}$ , then  $p_!$  preserves connected limits.*

*Proof.* By Remark 1.2.3 connected limits are computed in  $\mathcal{A}$  and  $p_!$  doesn't affect objects and morphism in  $\mathcal{A}$ .  $\square$

**Proposition 1.3.2.** *Let  $p : E \rightarrow B$  be a morphism in  $\mathcal{A}$ , then*

$$(\mathcal{A} \downarrow B) \quad \begin{array}{c} \xrightarrow{p^*} \\ \top \\ \xleftarrow{p!} \end{array} \quad (\mathcal{A} \downarrow E)$$

*is an adjunction.*

*In particular the unit is given by*

$$\begin{aligned} \eta : \text{Id}_{(\mathcal{A} \downarrow E)} &\rightarrow p^* p! \\ \eta_{(D, \delta)} = \langle \delta, 1_D \rangle : (D, \delta) &\rightarrow (E \times_B D, \text{proj}_1) \end{aligned}$$

$$\begin{array}{ccccc} D & & \xrightarrow{1_D} & & D \\ & \searrow \exists! \langle \delta, 1_D \rangle & & \searrow p\delta & \\ & E \times_B D & \xrightarrow{\quad} & D & \\ & \downarrow & & \downarrow & \\ & E & \xrightarrow{p} & B & \end{array}$$

*and the counit is given by*

$$\begin{aligned} \epsilon : p! p^* &\rightarrow \text{Id}_{(\mathcal{A} \downarrow B)} \\ \epsilon_{(A, \alpha)} = \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) &\rightarrow (A, \alpha) \end{aligned}$$

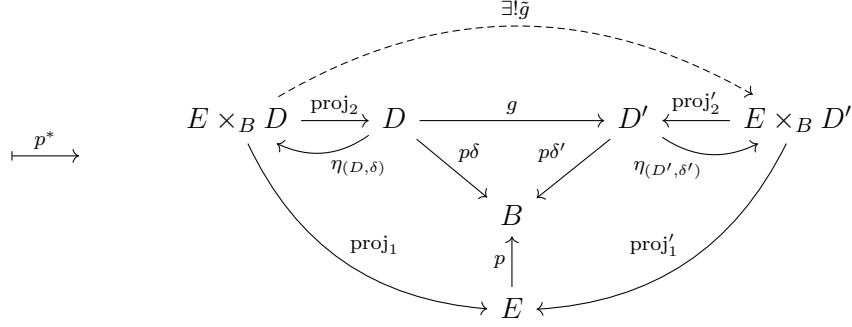
$$\begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_2} & A \\ \downarrow \text{proj}_1 & & \downarrow \alpha \\ E & \xrightarrow{p} & B. \end{array}$$

*Proof.* Let us show that this is actually an adjunction. First we want to prove that the unit is a natural transformation. Let  $g : (D, \delta) \rightarrow (D', \delta')$  be a morphism in  $(\mathcal{A} \downarrow E)$ , we want to prove that the following diagram commutes:

$$\begin{array}{ccc} D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D \\ \downarrow g & & \downarrow p^* p!(g) = \tilde{g} \\ D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D' \end{array}$$

In order to do that we need to expand the action of the functor  $p^* p!$  on the morphism  $g$ . We have:

$$\begin{array}{ccc} D & \xrightarrow{g} & D' \\ \downarrow \delta & \searrow \delta' & \\ E & & \end{array} \quad \xrightarrow{p!} \quad \begin{array}{ccc} D & \xrightarrow{g} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ & & B \end{array}$$



Considering the diagram

$$\begin{array}{ccccc}
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & & \\
 \downarrow g & \searrow \text{dashed} & \downarrow p^*p_!(g)=\tilde{g} & & \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
 & & \downarrow \text{proj}'_1 & & \downarrow p\delta' \\
 & & E & \xrightarrow{p} & B
 \end{array}$$

we can prove the naturality in the left diagram by the universal property of the pullback in the right. Since  $(\text{proj}_1, \text{proj}_2)$  are jointly monic, by the following compositions

- $(\text{proj}'_2)(\tilde{g})(\eta_{(D,\delta)}) = (g)(\text{proj}_2)(\eta_{(D,\delta)}) = (g)(1_D) = g$
- $(\text{proj}'_2)(\eta_{(D',\delta')})(g) = (1_{D'})(g) = g$
- $(\text{proj}'_1)(\tilde{g})(\eta_{(D,\delta)}) = (\text{proj}_1)(\eta_{(D,\delta)}) = \delta$
- $(\text{proj}'_1)(\eta_{(D',\delta')})(g) = (\delta')(f) = \delta$

we conclude that  $(\tilde{g})(\eta_{(D,\delta)}) = (\eta_{(D',\delta')})(g)$ . So the naturality condition of  $\eta$  is satisfied.

Now we want to prove that the counit is a natural transformation. Consider  $f : (A, \alpha) \rightarrow (A', \alpha')$  a morphism in  $(\mathcal{A} \downarrow B)$ , we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}} & A \\
 p_!p^*(f)=\tilde{f} \downarrow & & \downarrow f \\
 E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}} & A'
 \end{array}$$

As similarly done above, we need to expand the action of the functor  $p_!p^*$  on the morphism  $f$ , we have:

$$\begin{array}{ccc}
 \begin{array}{c} A \xrightarrow{f} A' \\ \downarrow \alpha \quad \swarrow \alpha' \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \curvearrowright & & \curvearrowleft & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \downarrow & \swarrow \alpha' & \\ & & B & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \searrow \text{proj}_1' \\ & E & & & \end{array} \end{array} \\
 & & \xrightarrow{p!} \\
 & & \begin{array}{c} \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \curvearrowright & & \curvearrowleft & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \downarrow & \swarrow \alpha' & \\ & & B & & \\ p(\text{proj}_1) \swarrow & & \uparrow & & \searrow p(\text{proj}_1') \\ & & & & \end{array} \end{array}
 \end{array}$$

Since  $\epsilon_{(A,\alpha)}$  is the  $\text{proj}_{2,E \times_B A}$ , the composition  $(f)(\epsilon_{(A,\alpha)}) = (\epsilon_{(A',\alpha')})(\tilde{f})$  holds by the definition of  $\tilde{f}$ . So also the naturality of  $\epsilon$  is satisfied.

Now we can check that the unit and counit fulfil the first triangle identities. Let  $(A, \alpha)$  be an object in  $(\mathcal{A} \downarrow B)$ , we have:

$$((p^* \epsilon) \circ (\eta p^*))_{(A,\alpha)} = (p^* \epsilon)_{(A,\alpha)} (\eta p^*)_{(A,\alpha)} = p^*(\text{proj}_{2,E \times_B A})(\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

Considering first how  $p^*$  acts on the morphism  $\text{proj}_{1,E \times_B A}$

$$\begin{array}{ccc}
 \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_{1,E \times_B A}} & A \\ & \searrow \alpha(\text{proj}_{2,E \times_B A}) & \downarrow \alpha \\ & & B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \text{proj}_{2,E \times_B A} \\ \curvearrowright \\ E \times_B (E \times_B A) \xrightarrow{\text{proj}_{2,E \times_B (E \times_B A)}} E \times_B A \xrightarrow{\text{proj}_{2,E \times_B A}} A \xleftarrow{\text{proj}_{2,E \times_B A}} E \times_B A \\ \quad \quad \quad \searrow (\alpha)(\text{proj}_{2,E \times_B A}) \quad \downarrow \alpha \quad \swarrow \text{proj}_{1,E \times_B A} \\ \quad \quad \quad \quad \quad \quad B \\ \quad \quad \quad \quad \quad \quad \uparrow p \\ \quad \quad \quad \quad \quad \quad E \end{array}
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\eta_{(E \times_B A, \text{proj}_1)})$$

and then considering the definition of  $\eta$ , we obtain

$$\begin{array}{ccccc}
 & & 1_{E \times_B A} & & \\
 & \nearrow & & \searrow & \\
 E \times_B A & & & & E \times_B A \\
 & \searrow \exists! \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle & & \nearrow & \\
 & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \\
 & \downarrow & & \downarrow p(\text{proj}_{1, E \times_B A}) & \\
 \text{proj}_{1, E \times_B A} \searrow & E & \xrightarrow{p} & B & \\
 & & & & 
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle)$$

In order to conclude we need to check some compositions so that, by the universal property of the pullback  $E \times_B A$  in the right of the following diagram, we obtain the identity morphism.

$$\begin{array}{ccccccc}
 & & 1_{E \times_B A} & & \tilde{\text{proj}}_2 & & \\
 & \nearrow & & \searrow & & \nearrow & \\
 E \times_B A & \xrightarrow{\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle} & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \xrightarrow{\text{proj}_{2, E \times_B A}} & A \longleftarrow E \times_B A \\
 & \searrow & \downarrow & & \downarrow (\alpha)(\text{proj}_{2, E \times_B A}) & \nearrow \alpha & \\
 & & & & B & & \\
 & \searrow \text{proj}_{1, E \times_B (E \times_B A)} & & & \uparrow p & \searrow \text{proj}_{1, E \times_B A} & \\
 & & E & & & & 
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle) = \text{Id}_{(E \times_B A, \text{proj}_1)}$$

We want to check that composition of the projections of  $E \times_B A$  with  $\circledast$  is the same of composing those projection with  $\text{Id}_{(E \times_B A, \text{proj}_1)}$ . Following the first composition

$$\begin{aligned}
 (\text{proj}_{2, E \times_B A})(\tilde{\text{proj}}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) &= (\text{proj}_{2, E \times_B A})(\text{proj}_{2, E \times_B (E \times_B A)})(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \\
 &= (\text{proj}_{2, E \times_B A})(1_{E \times_B A}) = \text{proj}_{2, E \times_B A} = (\text{proj}_{2, E \times_B A})(\text{Id}_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

and the second composition

$$\begin{aligned}
 (\text{proj}_{1, E \times_B A})(\tilde{\text{proj}}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) &= (\text{proj}_{1, E \times_B (E \times_B A)})(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \\
 \text{proj}_{1, E \times_B A} &= (\text{proj}_{1, E \times_B A})(\text{Id}_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

Finally we can check the second triangle identities. Let  $(D, \delta)$  be an object in  $(\mathcal{A} \downarrow E)$ , we have:

$$\begin{aligned} ((\epsilon p!) \circ (p! \eta))_{(D, \delta)} &= ((\epsilon p!)_{(D, \delta)} (p! \eta))_{(D, \delta)} = (\epsilon_{p!(D, \delta)} p! (\langle \delta, 1_D \rangle)) = \\ &= (\epsilon_{(D, p\delta)}) p! (\langle \delta, 1_D \rangle) = \text{proj}_{2, E \times_B D} (\langle \delta, 1_D \rangle) = 1_D = \text{Id}_{(D, \delta)} \end{aligned}$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_{2, E \times_B D}} & D \\ \downarrow \text{proj}_{1, E \times_B D} & & \downarrow p\delta \\ E & \xrightarrow{p} & B \end{array}$$

So we can conclude that the adjunction holds.  $\square$

**Remark 1.3.3.**  $p^*$  is a right adjoint functor, so it preserves limits.  $p_!$  is a left adjoint functor, so it preserves colimits.

We need the following definition in order to claim some properties.

**Definition 1.3.4.** Let  $\gamma$  be a natural transformation between two functors  $F, G : \mathcal{C} \rightarrow \mathcal{D}$ , we say that  $\gamma$  is a cartesian natural transformation if for every morphism  $f : A \rightarrow B$  in  $\mathcal{C}$ , the following diagram is a pullback:

$$\begin{array}{ccc} F(A) & \xrightarrow{\gamma_A} & G(A) \\ \downarrow F(f) & & \downarrow G(f) \\ F(B) & \xrightarrow{\gamma_B} & G(B) \end{array}$$

ie the naturality diagrams are pullback.

**Property 1.3.5.** The unit  $\eta$  and the counit  $\epsilon$  are cartesian.

*Proof.* First we prove that  $\eta$  is cartesian. Let  $g : (D, \delta) \rightarrow (D', \delta')$  be a morphism in  $(\mathcal{A} \downarrow E)$ . Considering the commutative diagram below, since both the bottom and the total diagrams are pullback, it follows that the top diagram is a pullback.

$$\begin{array}{ccccc} & E \times_B D & \xrightarrow{\text{proj}_2} & D & \\ & \downarrow p^* p!(g) = \tilde{g} & & \downarrow g & \\ \text{proj}_1 \swarrow & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' & \searrow p\delta \\ & \downarrow \text{proj}'_1 & & \downarrow p\delta' & \\ & E & \xrightarrow{p} & B & \end{array}$$

Considering the following commutative diagram, since the right diagram is a pullback, as we just proved, and the total diagram is a pullback, we can conclude that the left

diagram is a pullback, as stated.

$$\begin{array}{ccccc}
 & & 1_D & & \\
 & \nearrow & & \searrow & \\
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & \xrightarrow{\text{proj}_2} & D \\
 \downarrow g & & \downarrow p^*p_!(g)=\tilde{g} & & \downarrow g \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
 & \nwarrow & & \nearrow & \\
 & & 1_{D'} & & 
 \end{array}$$

Now we want to prove that  $\epsilon$  is cartesian. Let  $f : (A, \alpha) \rightarrow (A', \alpha')$  be a morphism in  $(\mathcal{A} \downarrow B)$ . Considering the following commutative diagram, since the bottom diagram is a pullback and the total diagram is a pullback, we can conclude that the top diagram is a pullback, as stated.

$$\begin{array}{ccccc}
 E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}=\text{proj}_2} & A & & \\
 \downarrow p_!p^*(f)=\tilde{f} & & \downarrow f & & \\
 E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}=\text{proj}'_2} & A' & & \\
 \downarrow \text{proj}'_1 & & \downarrow \alpha' & & \\
 E & \xrightarrow{p} & B & & 
 \end{array}$$

$\text{proj}_1$  (left curved arrow),  $\alpha$  (right curved arrow)

□

## 1.4 The monad associated to the adjunction

Consider the monad associated to the adjunction we just studied, we want to study some properties that will be useful.

**Property 1.4.1.** *The monad  $(p^*p_!, \eta, p^*\epsilon p_!)$  is cartesian.*

*Proof.* We need to prove that the category  $(\mathcal{A} \downarrow E)$  has pullback, the monad preserves pullbacks, and both the multiplication  $p^*\epsilon p_!$  and the unit  $\eta$  are cartesian.

The category  $(\mathcal{A} \downarrow E)$  has pullback, since pullbacks are connected limit and hence are computed in  $\mathcal{A}$ , which has pullbacks. The monad  $p^*p_!$  preserves pullbacks, in fact  $p^*$  is a right adjoint so it preserves limits, therefore preserves pullbacks, and  $p_!$  preserves pullbacks because (cita all'inizio). We already proved that the unit  $\eta$  is cartesian, so it remains to prove that the multiplication  $p^*\epsilon p_!$  is cartesian. Since  $\epsilon$  is cartesian and  $p^*$  preserves pullbacks as shown above, we can conclude that the multiplication is cartesian. □

**Definition 1.4.2.** *A monad  $(T, \eta, \mu)$  on a category  $\mathcal{X}$  with finite coproducts is said to be essentially nullary if the morphism  $[T(!_X), \eta_X]$  is a strong epimorphism for each object  $X$  in  $\mathcal{X}$ , where  $!_X$  is the unique morphism from the initial object of  $\mathcal{X}$  to  $X$ .*

Now we claim a lemma which will be useful to prove some theorems.

**Lemma 1.4.3.** *A category  $\mathcal{X}$  is protomodular id and only if for each pullback*

$$\begin{array}{ccc} B' \times_B A & \xrightarrow{k} & A \\ \downarrow \uparrow & & \downarrow \uparrow \\ B' & \xrightarrow{\beta} & B \end{array} \quad \begin{array}{c} p \\ s \end{array}$$

*the pair  $(k, s)$  is strongly epimorphic.*

The proof is out of the scope of this thesis, but we can find it in (citazione)

**Theorem 1.4.4.** *If  $\mathcal{X}$  is a pointed protomodular category with finite coproducts and  $(T, \eta, \mu)$  is a monad on  $\mathcal{X}$  with cartesian  $\eta$ , then the monad is essentially nullary.*

*Proof.* For each object  $X$  in  $\mathcal{X}$  we can consider the following diagram:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & T(X) \\ \downarrow \uparrow & & \downarrow \uparrow \\ 0 & \xrightarrow{\eta_0} & T(0) \end{array} \quad \begin{array}{c} !^X \\ T(!^X) \end{array}$$

Since  $\mathcal{X}$  is protomodular then by the lemma above, the pair  $(\eta_X, T(!^X))$  is a strongly epimorphic. It is a standard fact that given  $f$  and  $g$  morphism, if  $(f, g)$  is a strongly epimorphic pair, then  $[f, g]$  is a strong epi. So we conclude that  $[\eta_X, T(!^X)]$  is a strong epi, ie the moand is essentially nullary. (citazione)  $\square$

$$\begin{array}{ccccc} & C & & C' & \\ & \swarrow \exists! s & & \nwarrow \exists! t & \\ p^*(\text{Eq}(q)) & \xrightarrow[p^*(p_1)=\tilde{p}_1]{p^*(p_2)=\tilde{p}_2} & p^*(A) = A \times_B E & \xrightarrow{p^*(q)=\tilde{q}} & p^*(\text{Coeq}(p_1, p_2)) \\ \downarrow & & \downarrow & & \downarrow \\ \text{Eq}(q) & \xrightarrow[p_1]{p_2} & A & \xrightarrow{q} & \text{Coeq}(p_1, p_2) \end{array}$$